1 Introduction to Bayesian optimization

This chapter covers

- What motivates Bayesian optimization and how it works
- Real-life examples of Bayesian optimization problems
- A toy example of Bayesian optimization in action

I'm very happy that you are reading this book and excited for your upcoming journey. On a high level, Bayesian optimization is an optimization technique that may be applied when the function (or in general any process that gives you an output when an input is passed in) we are trying to optimize is a black box and expensive to evaluate in terms of time, money, or other resources. This setup encompasses many important tasks including hyperparameter tuning (which we will define shortly). Using Bayesian optimization could accelerate this search procedure and help us locate the optimum of the function as quickly as possible.

As a machine learning practitioner, you might have heard of the term Bayesian optimization from time to time, or you might never encounter it before. While Bayesian optimization has enjoyed enduring interest from the machine learning (ML) research community, it's not as commonly used and talked about as other ML topics in practice. Why? Some might say Bayesian optimization has a steep learning curve: you need to understand calculus, use some probability, and overall be an experienced ML researcher to use Bayesian optimization in your application. Our goal for this book is to dispel the message that Bayesian optimization is difficult to use, and show that the technology is more intuitive and accessible than one would think.

Throughout this book, we will see a lot of illustrations, plots, and of course, code, which will help make whichever the topic currently being discussed more straightforward and concrete. You will learn how each component of Bayesian optimization works on a high level and how to implement them using state-of-the-art libraries in Python. Another hope of mine for the

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